Train a Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One complete classifier experiment record.

Trace stored values, raw queries, predictions and matched baseline errors.

Keep one habit: W1–W3/W5 → selected P/Q/R run → project experiment record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Nearest Neighbors

> **My remembered idea, example and correction**

Baselines First

> **My remembered idea, example and correction**

Sampling and Intended Use

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A(10,200)clear, B(20,100)blocked, C(20,200)blocked; ordered features distance cm/brightness units. Fit minima(10,100),ranges(10,100),scaled A(0,1),B(1,0),C(1,1). Raw V1(11,120),V2(19,190),V3(12,80) scale to(0.1,0.2),(0.9,0.9),(0.2,-0.2), select A/C/B and predict clear/blocked/blocked. References clear/blocked/clear give2/3,majority-blocked baseline1/3. V3 remains wrong on identical reruns.

  Fixed cm→mm conversion in the separate picture multiplies by10; it does not learn a scale.

- List what fit learns or stores, what **predict** takes as **input**, and what evaluation compares. State where the validation reference labels belong.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A(10,200)clear, B(20,100)blocked, C(20,200)blocked; ordered features distance cm/brightness units. Fit minima(10,100),ranges(10,100),scaled A(0,1),B(1,0),C(1,1). Raw V1(11,120),V2(19,190),V3(12,80) scale to(0.1,0.2),(0.9,0.9),(0.2,-0.2), select A/C/B and predict clear/blocked/blocked. References clear/blocked/clear give2/3,majority-blocked baseline1/3. V3 remains wrong on identical reruns.

  Fixed cm→mm conversion in the separate picture multiplies by10; it does not learn a scale.

- Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the **training**-majority baseline. Which error should repeat in an unchanged rerun?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A(10,200)clear, B(20,100)blocked, C(20,200)blocked; ordered features distance cm/brightness units. Fit minima(10,100),ranges(10,100),scaled A(0,1),B(1,0),C(1,1). Raw V1(11,120),V2(19,190),V3(12,80) scale to(0.1,0.2),(0.9,0.9),(0.2,-0.2), select A/C/B and predict clear/blocked/blocked. References clear/blocked/clear give2/3,majority-blocked baseline1/3. V3 remains wrong on identical reruns.

  Fixed cm→mm conversion in the separate picture multiplies by10; it does not learn a scale.

- A classmate saves only k=1 and a score of 2/3. Name the information needed to **reproduce** this run. Explain why the same score alone is insufficient.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training A(10,200)clear, B(20,100)blocked, C(20,200)blocked; ordered features distance cm/brightness units. Fit minima(10,100),ranges(10,100),scaled A(0,1),B(1,0),C(1,1). Raw V1(11,120),V2(19,190),V3(12,80) scale to(0.1,0.2),(0.9,0.9),(0.2,-0.2), select A/C/B and predict clear/blocked/blocked. References clear/blocked/clear give2/3,majority-blocked baseline1/3. V3 remains wrong on identical reruns.

  Fixed cm→mm conversion in the separate picture multiplies by10; it does not learn a scale.

- Explain and repair three mistakes: **predict** receives an already scaled query; fit receives **training** plus validation rows; a query uses brightness before distance. Does repeating each mistaken run make it valid?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This new task uses training P/Q/R and its own reference labels, not A/B/C. Fit only those supplied training rows, use raw feature pairs in the same distance/brightness order, and keep the resulting state for all three queries. Reusing the earlier fitted A/B/C state would answer a different task. No final test or broad reliability evidence is supplied.

- Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

  Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Build and trace a fresh fitted state

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Trace and score the saved run for A reproducible classifier **pipeline**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: “A **pipeline** prevents every leak automatically”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- **Training**: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).
- Apply the rule and distinguish **fitting** from inference.
- Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional **evidence** is needed before claiming generalization?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 13 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training**: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Pipeline rules: fit each feature's minimum and range (maximum minus minimum) using training rows only. Scale every training and query component as (value – fitted minimum) / fitted range, without clipping. For k=1, sum absolute component differences and break distance ties by alphabetical ID. The baseline is the training-majority label. Do not refit on validation queries.

- Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

  Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

My reasoning and evidence

▼ **Extension · Searching candidate rules — W1 State the search before running it**

REFERENCE · SECTION 17

Extension · Searching candidate rules — W1 State the search before running it

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Write the rule direction, equality behavior, candidate cutoffs and tie policy. Identify what humans supplied and what **fitting** selects.

My reasoning and evidence

▼ Extension · Searching candidate rules — W2 Trace the training mistakes

REFERENCE · SECTION 18

Extension · Searching candidate rules — W2 Trace the training mistakes

> Retained extension task with its own values and teacher key.

Extension question and evidence

- List errors for cutoffs 3, 5 and 7 on the worked **training** rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

> **My reasoning and evidence**

▼ Extension · Searching candidate rules — W3 Audit the boundary and comparison

REFERENCE · SECTION 19

Extension · Searching candidate rules — W3 Audit the boundary and comparison

> Retained extension task with its own values and teacher key.

Extension question and evidence

- What happens at distance 5 if **prediction** uses < instead of <= with cutoff 5? Explain why the search uses a strict < for error counts even though **prediction** uses <= for distance.

> **My reasoning and evidence**

▼ **Extension · Searching candidate rules — W4 Search and evaluate independently**

REFERENCE · SECTION 20

Extension · Searching candidate rules — W4 Search and evaluate independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- New **training** rows are (1,blocked), (3,clear), (5,blocked), (7,clear). Try cutoffs 2,4,6, using the same direction and smaller-cutoff tie policy. Give every **training error** count and select a cutoff.

  Validation rows are (2,blocked), (4,blocked), (6,clear), (8,clear). List the selected rule's **prediction**s and score. Compare with the **training**-majority baseline using clear on ties.

My reasoning and evidence

▼ **Extension · Searching candidate rules — W5 Limit the claim**

REFERENCE · SECTION 21

Extension · Searching candidate rules — W5 Limit the claim

> Retained extension task with its own values and teacher key.

Extension question and evidence

- The worked rule fits **training** perfectly but ties the baseline on validation accuracy. Explain why neither result proves it is the best possible model. Name two details to keep in the experiment record.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.
- Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance. Query is 5 cm.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/13

Train a Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit transform

The declared conversion produces training lengths 20 and 60 mm.

↓

2. Transform query

5 cm becomes 50 mm using the same conversion.

↓

3. Predict neighbour

Distances 30 and 10 mm select the spotted training record.

CHANGE ONE THING

Transform training but leave query numerically at 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.
- Which labelled step matches this detail? Distances 30 and 10 mm select the spotted **training** record.
- 1. **Predict** neighbour
- 2. Fit transform
- 3. Transform query
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Predict** neighbour Distances 30 and 10 mm select the spotted **training** record. Read the picture in order. Distances 30 and 10 mm select the spotted **training** record.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Transform **training** but leave query numerically at 5. Which conclusion is supported by the supplied facts?
- 1. Distances become 15 and 55, incorrectly selecting healthy.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Repair and keep the record

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Define the population and conditions in an intended-use statement. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. **Predict** first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Train a Classifier: the case

AI ACADEMY / AI-4/13 · 01 · LOOK AT THE FACTS

Train a Classifier: the case

The situation

Separate a reproducible fitted pipeline from merely displaying predictions.

Supplied evidence

Toy training lengths are [2, 6] cm with labels [healthy, spotted]. Pipeline converts cm to mm, then uses k=1 absolute distance.
Query is 5 cm.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain train a classifier using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

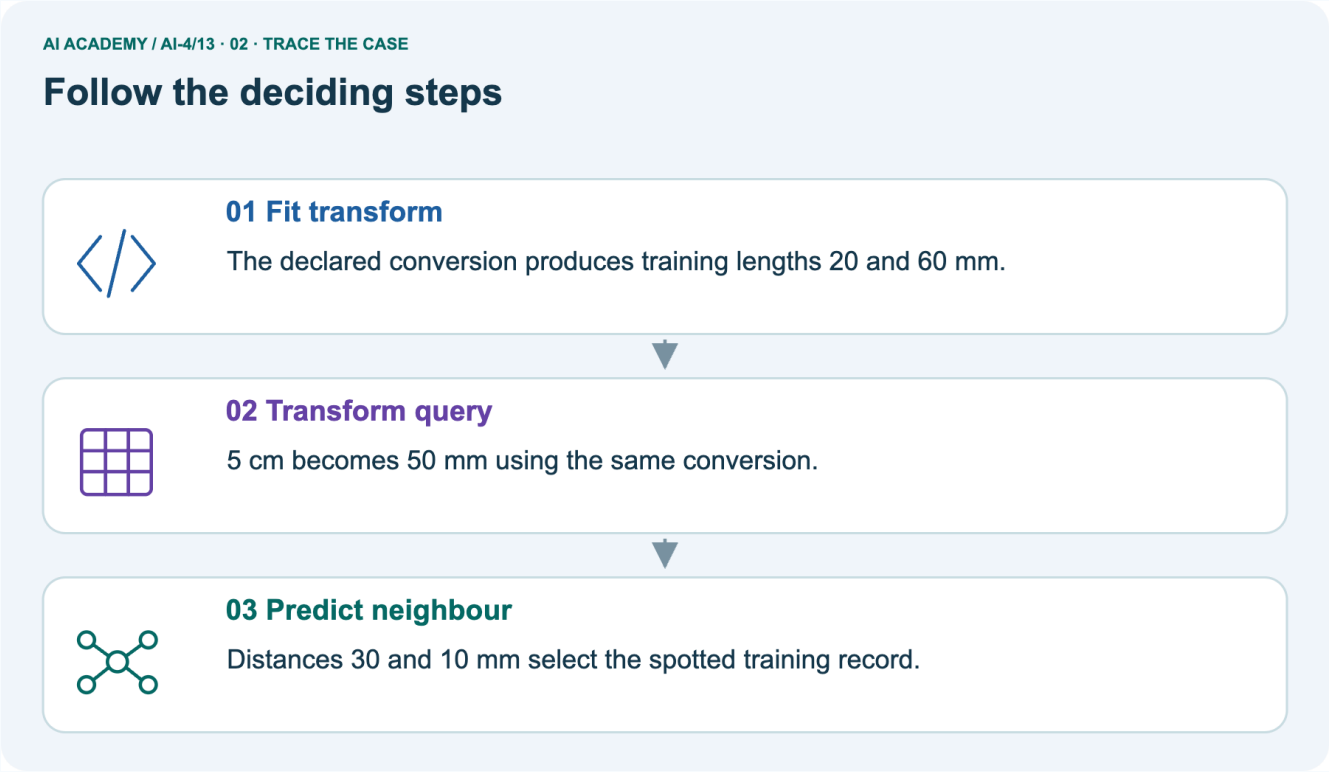
Separate a reproducible fitted **pipeline** from merely displaying **prediction**s. Supplied facts: Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance.

Query is 5 cm. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



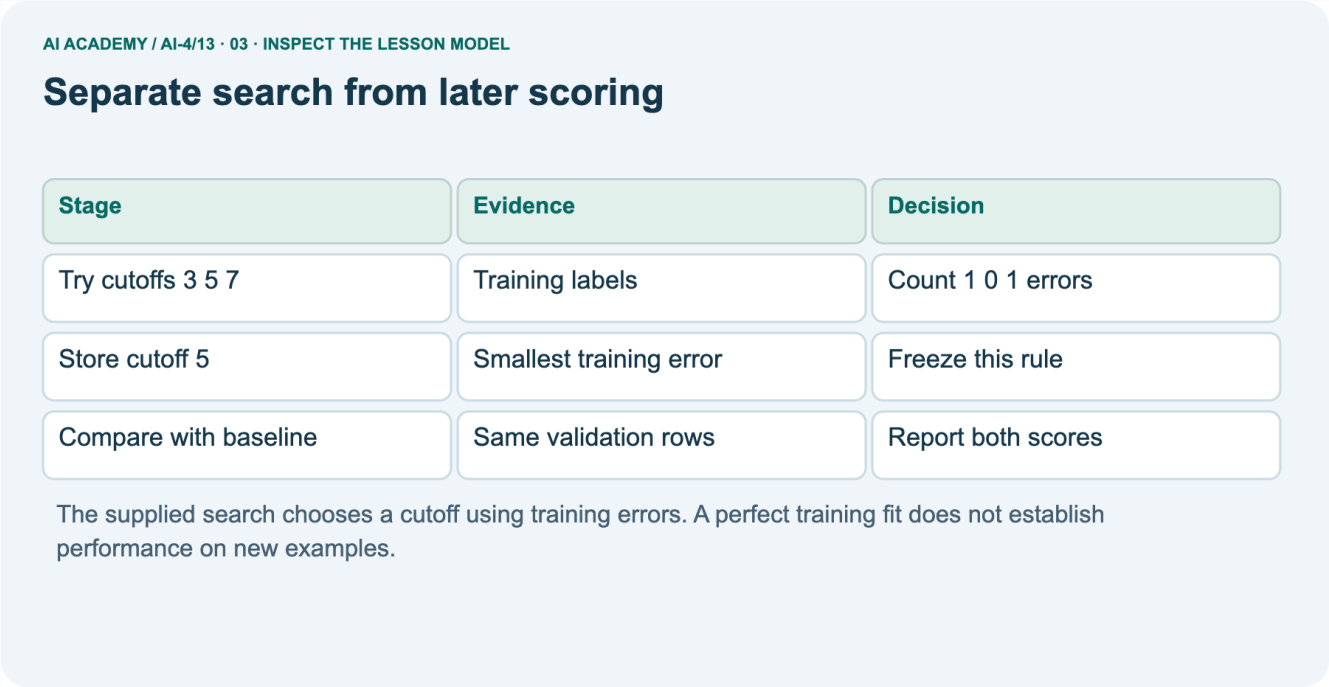
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit transform: The declared conversion produces **training** lengths 20 and 60 mm. → Transform query: 5 cm becomes 50 mm using the same conversion. → **Predict** neighbour: Distances 30 and 10 mm select the spotted **training** record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Separate search from later scoring



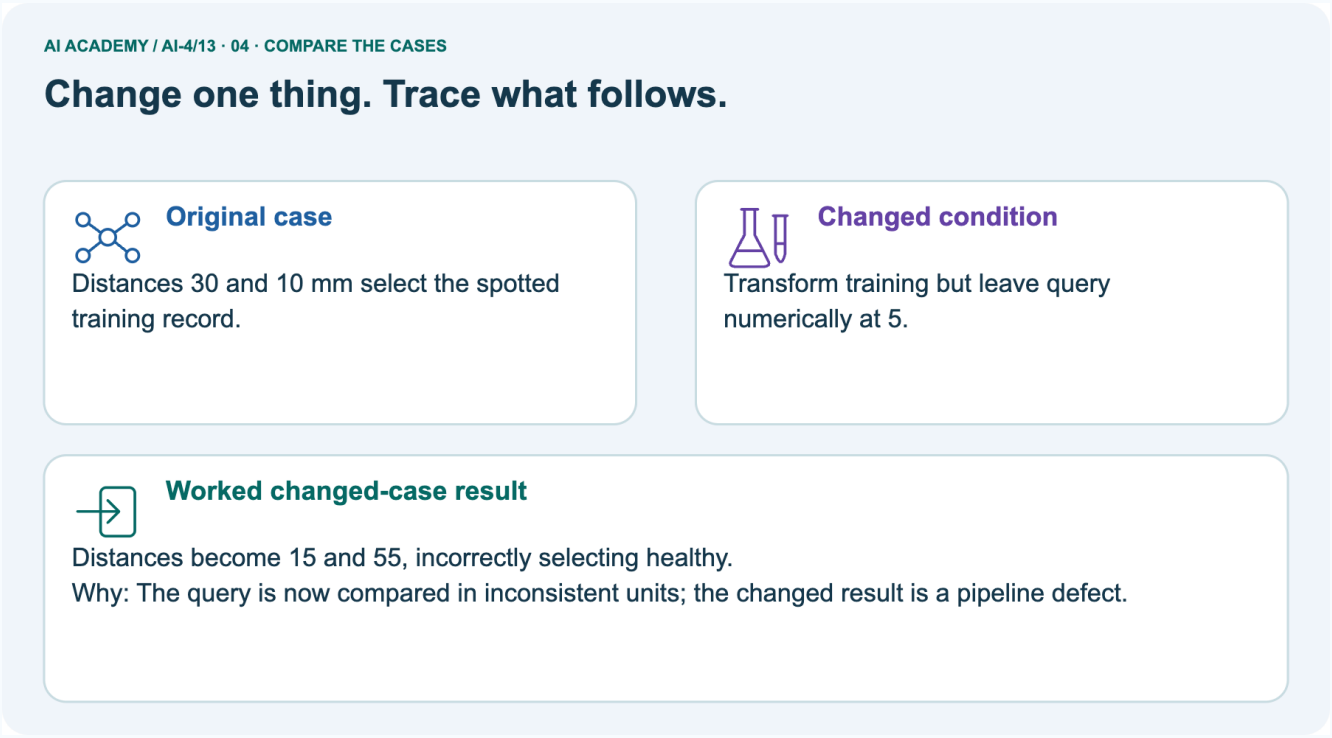
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate search from later scoring. Try cutoffs 3 5 7 | **Training** labels | Count 1 0 1 errors. Store cutoff 5 | Smallest **training error** | Freeze this rule. Compare with baseline | Same validation rows | Report both scores

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

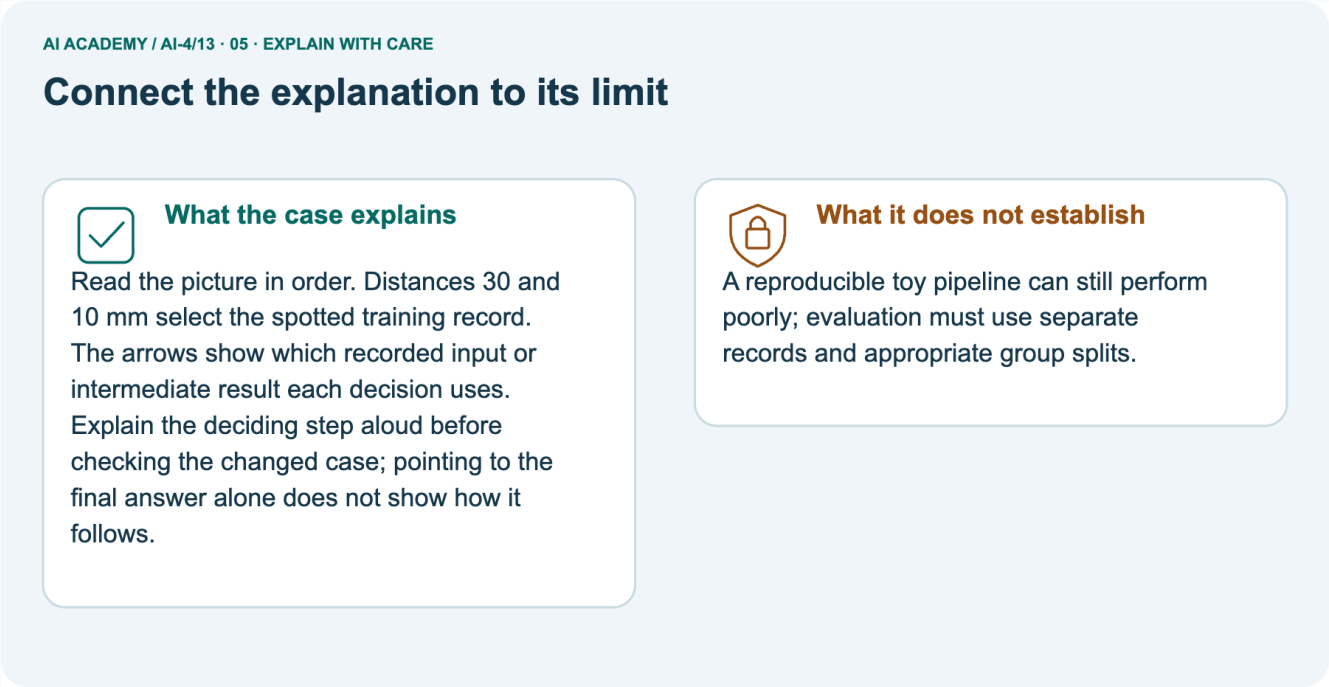
Original case: Distances 30 and 10 mm select the spotted **training** record. Changed condition: Transform **training** but leave query numerically at 5. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

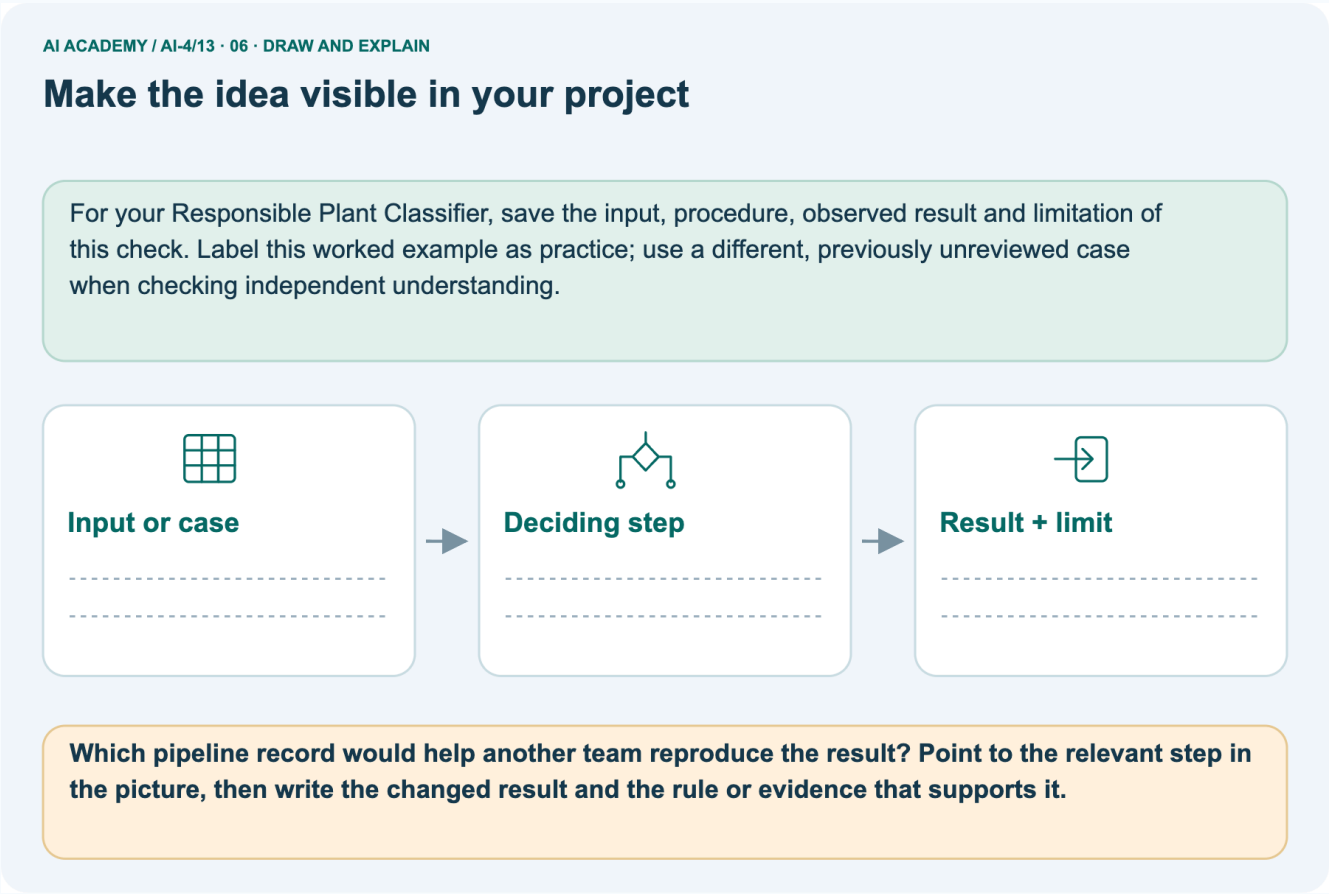
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Train a Classifier: the case

02. Follow the deciding steps

03. Separate search from later scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.